

APPLICATIONS

CHANGE-POINT ANALYSIS OF THE CD4 CELL COUNT IN HIV-POSITIVE PATIENTS

Pieter C.N. Groenewald

Department of Mathematical Statistics, University of the Free State, South Africa.

e-mail: *groenepc@sci.uovs.ac.za*

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Summary: The CD4 cell count is sometimes used as a surrogate marker for the diagnosis of acquired immune deficiency syndrome (AIDS) in persons with the human immunodeficiency virus (HIV). However, a number of researchers have questioned the use of absolute CD4 cell counts and have suggested that the rate of decline in CD4 cells may be a better marker for the progression to AIDS. In this paper the CD4 cell count sequences of 83 patients from ACTG 175, receiving one of four different treatments, are analysed using a hierarchical Bayes change-point model. The purpose is to determine whether there is a sudden decline in cell count some time before diagnosis of AIDS and to find the posterior distribution of the mean time from change-point to diagnosis of AIDS. The fractional Bayes factor is used to calculate the probability of a change-point in an individual sequence, and censored observations are included. There is evidence that in most cases the diagnosis of AIDS is preceded by a sudden increase in the rate of decline in the number of CD4 cells. Results indicate that the time from change-point to AIDS diagnosis has a mean of 381 days and a standard deviation of 177 days.

1. Introduction

Since the study by Eyster et al. (1985), and through numerous subsequent studies, it has been established that the number of CD4 cells in the blood (per

mm^3) is an important biological marker for the disease progression in persons with the human immunodeficiency virus (HIV). The HIV selectively infects and destroys CD4 cells, which play an important role in the human immunity system. The level and slope of the CD4 cell counts over time are closely linked with the incidence of clinical manifestations of HIV and is sometimes used as a surrogate marker for the diagnosis of acquired immune deficiency syndrome (AIDS) (a CD4 count below 200).

However, a number of researchers have questioned the use of CD4 cell counts as a marker for AIDS. See, for example, Lin et al. (1993), Choi et al. (1993) and Lange, Carlin and Gelfand (1992). In these studies it was found that CD4 cell count alone is not an adequate marker for some of the other criteria by which a person is diagnosed as having AIDS, such as Kaposi's sarcoma, opportunistic infection or lymphoma.

Where the absolute number of CD4 cells may not be adequate, the rate of decrease in CD4 counts may be an indication of imminent onset of AIDS. Burcham et al. (1991), in a study with HIV-positive homosexual men, observed that the rate of decline in percent CD4 cells for subjects who developed AIDS showed a steeper decline than for those who did not develop AIDS. Eyster et al. (1989) analysed data from 84 haemophiliacs and found an abrupt decrease in cell count in most patients who developed AIDS at one to two years before diagnosis. Lang et al. (1989) also noted that among seropositive men who developed AIDS, the mean CD4 level fell rapidly during the 18 months prior to diagnosis.

The above-mentioned studies suggest a change-point model for CD4 cell counts prior to AIDS, and Lange et al. (1992) modelled the square root of CD4 count by a hierarchical Bayes change-point model, using data from the San

Francisco Men's Health Study on 327 HIV-positive men. They used a piece-wise linear model and incorporated features such as unknown seroconversion times, random change-points, heterogeneous variances and covariates. They concluded, however, that the data did not support a change-point model adequately. Kiuchi et al. (1995) analysed data from 131 homosexual men who entered the New York Blood Center Study in 1984 and who developed AIDS as participants by 1990. The log(CD4) count was modelled as a piece-wise linear model and empirical and hierarchical Bayes methods were used to estimate the distribution of the time before AIDS when a rapid decline in cell count begins. They found significant evidence for a change-point model with mean change-point approximately one year before AIDS diagnosis, with standard deviation of 9 months. In Carlin and Louis (2000), data from a trial involving 467 persons with advanced HIV infection are analysed. They used a mixed effect change-point model with fixed known change-point on the square root of the CD4 cell count, and the main aim of the analysis was to compare the effects of two antiretroviral drugs, didanosine and zalcitabine, on CD4 count, survival time and AIDS diagnosis.

Other approaches to the analysis of multi-path change-point data include Joseph and Wolfson (1992), Joseph, Vandal and Wolfson (1996), Joseph et al. (1997), BuHarma (1997) and Bélisle et al. (1998). Recently Bélisle et al. (2002) used a Bayesian change-point model to model the cognitive decline of patients with Alzheimer's disease. In Asgharian and Wolfson (2001) a model is proposed for the inclusion of covariates in the change-point distribution. This was done by modelling the change-point problem as a Markov chain and then to introduce the covariates through the hazard function.

In this paper we model the log(CD4) count by means of a hierarchical Bayes change-point model. The data is from the AIDS Clinical Trial Group

Study Protocol 175 (ACTG 175) and is described in the next section. The main purpose of this analysis is to determine if there are any differences in the effect of four antiretroviral treatments on the profiles of the CD4 cell counts and to find the posterior distribution of mean time from the change-point to the diagnosis of AIDS. So our aims are similar to those of Kiuchi et al. (1995), but with three important generalisations to the model. Firstly, as not all patients who developed AIDS exhibit an obvious change-point in their sequences of CD4 counts, the uncertainty regarding the existence of a change-point for each sequence is incorporated into the analysis. Secondly, censored observations are included. That is, a patient who has not yet developed AIDS, but whose sequence of CD4 counts shows evidence of a change-point, is included in the analysis. Thirdly, the effects of treatments are examined by incorporating them as covariates into the model.

2. Data

AIDS Clinical Trial Group 175 (ATCG 175) was a randomised double-blind placebo-controlled trial involving 2467 HIV-infected patients, who were randomly assigned to one of four antiretroviral treatments: zidovudine alone (AZT), didanosine alone (DDI), zidovudine plus didanosine (AZT+DDI), or zidovudine plus zalcitabine (AZT+DDC). Patients were recruited from AIDS Clinical Trial Units in the United States and Puerto Rico. The details and results of this study are given in Hammer et al. (1996). A subgroup of 391 patients, each of whom entered the study HIV-infected but AIDS-free, and with a CD4 cell count of between 200 and 500 was the subject of a study by Katzenstein et al. (1996). Of the 391 patients, 32 developed AIDS during the study period. Note that the diagnosis of AIDS was based on several criteria, as mentioned in the introduction, and not on the CD4 counts. Of the rest of the

patients, 51 showed some evidence of a change-point in the sequence of CD4 counts according to the criterion to be discussed in the next section. Patients entered the study at various times, but all censoring occurred at the end of the study period. The data that will be considered for our analysis consist of these 83 sequences, 51 of which are censored. The numbers of observations per treatment are as follows:

	AZT	DDI	AZT+DDI	AZT+DDC
Uncensored	11	9	5	7
Censored	12	13	11	15

The study ran for about three years, from 1992 to the beginning of 1995, with CD4 cell counts taken, on the average, every three months, although there can be considerable variation in interval lengths. The number of observations per subject varies between 5 and 15. Figure 1 shows the plots of the $\log(\text{CD4})$ counts over time for the 32 AIDS-diagnosed patients in (a), and for the 51 censored observations in (b) for all treatments. Since time of seroconversion is unknown and patients entered the study at various stages of HIV progression, time in (a) is rescaled with the origin at time of AIDS diagnosis, while in (b) the origin is at time of censoring.

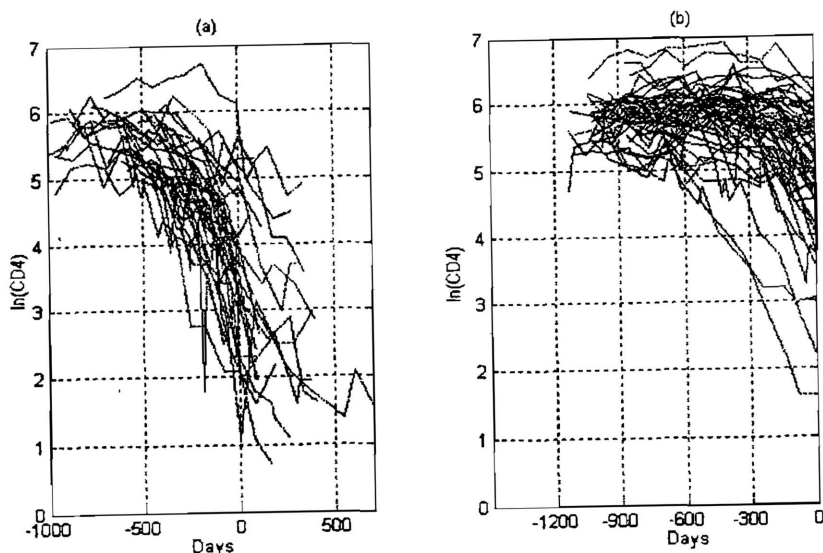


Figure 1: (a) Sequences of $\log(\text{CD4})$ cell counts of 32 patients who developed AIDS (origin).
 (b) Sequences of $\log(\text{CD4})$ cell counts of 51 patients who were HIV-positive but still AIDS-free (origin at time of censoring).

3. Probability of a change-point in a single sequence

As there is uncertainty about the existence of a change-point in each sequence of $\log(\text{CD4})$ cell counts, let M_k , $k = 2, 3, \dots, n-1$, denote the piece-wise linear model with exactly one change-point in a sequence, $y_1, y_2, \dots, y_n$, observed at time-points $x_1 < x_2 < \dots < x_n$, where the change-point, x_0 , lies in the interval $(x_k, x_{k+1}]$. Then

$$y_i = \alpha + \beta x_i + \tau(x_i - x_0)^+ + \varepsilon_i, \quad i = 1, 2, \dots, n, \quad (3.1)$$

where $c^+ = \max(0, c)$, and $x_k < x_0 \leq x_{k+1}$. The parameters α and β are the intercept and slope of the first line segment, while τ is the change in slope at the change-point. We assume further that $\varepsilon_i \sim \text{iid } N(0, \sigma^2)$, $i = 1, 2, \dots, n$.

In vector notation the observation model can then be written as

$$y|M_k \sim N(X_k\gamma, \sigma^2 I_n), \quad (3.2)$$

where $X_k(n \times 3) = [\mathbf{1}_n, \mathbf{x}, \mathbf{x}_k]$, $\gamma = [\alpha, \beta, \tau]'$, $\mathbf{1}_n$ is a vector of ones and the i -th element of $\mathbf{x}_k$ is $(x_i - x_0)^+$. Let M_0 denote the model with no change-point, i.e.

$$y|M_0 \sim N(X\beta, \sigma_0^2 I_n), \quad (3.3)$$

where $\beta = [\alpha_0, \beta_0]'$ and $X(n \times 1) = [\mathbf{1}_n, \mathbf{x}]$.

We will use fractional Bayes factors to determine the posterior probabilities of the change-point model for each sequence. In the absence of proper priors, Bayes factors are indeterminable due to the lack of normalizing constants, especially when the competing models have parameter spaces of different dimensions. A solution to this problem is what is generally called partial Bayes factors, by which is meant that part of the information from the data is combined with improper priors to produce proper ones. The Bayes factor is then computed using the rest of the data. A number of variations have been suggested, such as the imaginary training sample method of Spiegelhalter and Smith (1982), the intrinsic Bayes factor of Berger and Pericchi (1995, 1996) and the fractional Bayes factor of O'Hagan (1995, 1997). A general discussion of Bayes factors can be found in Kass and Raftery (1995), and a comparison of partial Bayes factors is done in Berger and Pericchi (1998). Dey and Purkayastha (1997) use the imaginary training sample method approach in change-point analysis.

The fractional Bayes factor uses a fraction $b < 1$ of the likelihood together with a vague prior to construct a proper prior under a specific model. Let

$$m_j(y) = \int (y|M_j, \theta_j) \pi(\theta_j) d\theta_j$$

be the marginal distribution of the data under model M_j with parameters θ_j , and

$$m_j^b(y) = \int [f(y|M_j, \theta_j)]^b \pi(\theta_j) d\theta_j$$

the fractional marginal. Then the fractional Bayes factor (FBF) in favour of model M_j , when compared with model M_i , is given by

$$B_{ji} = \frac{m_j(y) m_i^b(y)}{m_j^b(y) m_i(y)} \quad (3.4)$$

The posterior probability for model M_j out of m possible models is then

$$P(M_j|y) = P(M_j) \left[\sum_{i=1}^m B_{ij} P(M_i) \right]^{-1}, \quad (3.5)$$

where $P(M_j)$ is the prior probability assigned to model M_j . The suggested value of b (O'Hagan, 1995) is the smallest fraction of the sample necessary to keep $m_j^b(y)$ finite for all models. For this change-point model it means that $b = 4/n$.

If M_c denotes the model with a change-point, not conditional on k , then M_c is the union of models M_2 to M_{n-1} . The posterior probability of no change-point then follows from (3.5) as

$$P(M_0|y) = \left[1 + \frac{1}{B_{0c}} \right]^{-1} \quad (3.6)$$

with equal prior weights for the two models. With vague improper priors $\pi(\gamma) \propto 1$, $\pi(\beta) \propto 1$, $\pi(\sigma^2) \propto 1/\sigma^2$ and $\pi(\sigma_0^2) \propto 1/\sigma_0^2$, and with a uniform prior on x_0 between x_2 and x_{n-1} , all independently distributed, it

follows that

$$B_{0c} = \frac{\sqrt{\pi} \Gamma\left(\frac{n-2}{2}\right)}{\Gamma\left(\frac{n-3}{2}\right)} \frac{Q_1}{Q_{n-3}} [y'Hy]^{-\frac{n-4}{2}}, \quad (3.7)$$

where

$$Q_a = \sum_{k=2}^{n-2} \int_{x_k}^{x_{k+1}} (x'_k H x_k)^{-\frac{1}{2}} [y' A y]^{-\frac{a}{2}} dx_0 \quad (3.8)$$

and

$$H = I_n - X(X'X)^{-1}X',$$

$$A = H - Hx_k(x'_k H x_k)^{-1}x'_k H. \quad (3.9)$$

The posterior probability of a change-point then follows from (3.6).

Under the assumption of a change-point, the marginal posterior distribution of x_0 is given by

$$\pi(x_0|y) \propto (x'_k H x_k)^{-\frac{1}{2}} [x'Hy]^{-\frac{n-3}{2}} \quad \text{for } x_2 \leq x_0 < x_{n-1}. \quad (3.10)$$

Note that k is the largest integer smaller than or equal to x_0 , so the density function has discontinuities at the observed x -values. The posterior of τ conditional on x_0 is a t -distribution with mean $(x'_k H x_k)^{-1}y' H x_k$ and precision $(n-3)x'_k H x_k (y' A y)^{-1}$. The conditional posterior of σ^2 is an inverse Gamma distribution with parameters $n-3$ and $\frac{1}{2}y' A y$.

Applying the FBF to the log(CD4) cell counts of the 32 patients from ACTG175 that developed AIDS, and with equal prior probabilities, the posterior probabilities for a change-point vary between 0.243 and 0.999 with an average of 0.670. Assuming a change-point, the posterior mean of τ shows an increase in the rate of decline in CD4 cells in all but one of the cases. These posterior probabilities can also be considered as a goodness-of-fit measure relative to a linear model. The sequences of the rest of the patients were tested for change-points and those retained that showed some evidence (a posterior probability of larger than 0.5) of a change-point and with a negative posterior expectation of τ , indicating a downward change in slope. There

were 51 cases, which constitute the sample of censored observations. For this set, the posterior probabilities for a change vary between 0.51 and 0.998, with an average of 0.768.

4. Hierarchical Bayes Model

In the full hierarchical model we assume an underlying distribution for the position of a change-point, with common variance but with mean depending on the treatment. In Kiuchi et al. (1995) a normal prior is assumed for the change-point in log(CD4) counts. In Joseph and Wolfson (1992), a nonparametric empirical Bayes prior was used in the case of discrete change-point, while Joseph et al. (1970) and Bélisle et al. (1998) use noninformative Dirichlet priors for the multinomial probabilities of discrete change-points. Lange, Carlin and Gelfand (1992) use a discrete uniform prior over the period from 6 to 18 months after seroconversion. We have considered a lifetime distribution, such as the Weibull, for the time from change-point to diagnosis of AIDS, but since there is a positive probability for a change after diagnosis, we are assuming a normal distribution.

With the x-axis scaled so that time of AIDS diagnosis or time of censoring is at zero, let x_{0j} denote the position of the change-point for sequence j , $j = 1, 2, \dots, m$, assuming a change-point. Then

$$-x_{0j}|\eta, \phi^2 \sim \text{iid } N(z_j\eta, \phi^2), \quad j = 1, 2, \dots, m, \quad (4.1)$$

where $z_j(1 \times 4)$ indicates the treatment of the j -th patient and $\eta(4 \times 1)$ are the regression coefficients. We also have a vague uniform second-stage prior on η and an inverse Gamma prior with parameters (a, d) on ϕ^2 . The reason for a prior on the negative of x_{0j} is mainly for convenience. Now we are working with right-censored observations and $z_j\eta$, the mean time from change-point to AIDS, is positive.

For the regression parameters of the j -th subject, $\gamma_j = [\alpha_j, \beta_j, \tau_j]'$, we have

$$\gamma_j \sim N(Mz'_j, \Sigma), \quad j = 1, 2, \dots, m, \quad (4.2)$$

with vague uniform priors on the elements of $M(3 \times 4)$ and $\Sigma^{-1} \sim W((\rho\Phi)^{-1}, \rho)$, a Wishart distribution with expectation Φ^{-1} and ρ degrees of freedom.

Under the assumption of a change-point, let $m = m_1 + m_2$, where m_1 is the number of complete observations and m_2 the number of censored observations. With X_{kj} as defined in section 3, the full conditional distributions of all parameters are as follows:

$$1. f(x_{oj}|y_j, \gamma_j, \sigma^2, \eta, \phi^2) \propto e^{-\frac{1}{2\sigma^2} (y_j - X_{kj}\gamma_j)'(y_j - X_{kj}\gamma_j) - \frac{1}{2\phi^2} (x_{oj} + z_j\eta)^2}$$

for $j = 1, 2, \dots, m_1$. (4.3)

$$1a. f(x_{oj}|y_j, \gamma_j, \sigma^2, \eta, \phi^2) \propto e^{-\frac{1}{2\sigma^2} (y_j - X_{kj}\gamma_j)'(y_j - X_{kj}\gamma_j)} \\ \times S(-x_{oj}; z_j\eta, \phi^2) \quad \text{for } j = 1, 2, \dots, m_2. \quad (4.3a)$$

$$2. \gamma_j|y_j, M, \sigma^2, \Sigma, x_{oj} \sim N \left[(X'_{jk}X_{kj} + \sigma^2\Sigma^{-1})^{-1} \right. \\ \left. \times (X_{kj}y_j + \sigma^2\Sigma^{-1}Mz'_j); \sigma^2(X'_{kj}X_{kj} + \sigma^2\Sigma^{-1})^{-1} \right] \\ \text{for } j = 1, 2, \dots, m. \quad (4.4)$$

$$3. f(\eta, \phi|x_{oj}, j = 1, \dots, m) \propto (\phi^2)^{-(m_1/2 + a + 1)} \\ \times e^{-\frac{1}{\phi^2} [d + 1/2 \sum_{j=1}^{m_1} (x_{oj} + z_j\eta)^2]} \prod_{j=1}^{m_2} S(-x_{oj}; z_j\eta, \phi^2)$$

where $S(\cdot; \cdot)$ is the survival function of a normal variate. (4.5)

$$4. \sigma^2|y_j, \gamma_j, x_{oj} \sim IG \left(\frac{1}{2} \sum_{j=1}^m n_j, \frac{1}{2} \sum_{j=1}^m (y_j - X_{kj}\gamma_j)'(y_j - X_{kj}\gamma_j) \right) \quad (4.6)$$

$$5. M|\gamma_j, \Sigma \sim N(\gamma Z'(ZZ')^{-1}, (ZZ')^{-1} \otimes \Sigma)$$

where $Z = [z'_1, z'_2, \dots, z'_m]$ is $(m \times 4)$ and γ is $(3 \times m)$. (4.7)

$$6. \Sigma^{-1} | \gamma_j, M \sim W \left[\left\{ \rho \Phi + \sum_{j=1}^m (\gamma_j - M z'_j)(\gamma_j - M z'_j) \right\}^{-1}, m + \rho \right] \quad (4.8)$$

Notice that censoring only plays an explicit role in equations (4.3a) and (4.5), the joint conditional distribution of the mean and variance of the time between the change-point and diagnosis. Apart from equations (4.3), (4.3a) and (4.5), all conditional distributions are standard and the Gibbs sampler was used to generate parameter values from these distributions. The Metropolis-Hastings algorithm was used to generate from (4.3), (4.3a) and (4.5).

As there is uncertainty about the existence of a change-point in any sequence, let $p_j = P[M_c^{(j)} | y_j]$ be the probability of a change-point in sequence j , $j = 1, \dots, m$, where p_j is calculated as in section 3. Then each of the generated sets of x_{0j} and γ_j from (4.3) and (4.4) above is used in the generation of the other five parameters with probability p_j . This is done at the beginning of each cycle of the sampling scheme so that the eventual weight assigned to each sequence in estimating the parameters in (4.4) to (4.7) is proportional to their respective probabilities of having a change-point. So m in equations (4.4) to (4.8) does not refer to the total number of subjects, but to the number included in the analysis during each cycle.

5. Results

For the known hyperparameters, $\Phi = [1.5 \quad 0.00002 \quad 0.00002]$ was used. These values are approximately the moments estimators of the diagonal elements of Σ , but little weight is attached to Φ by taking $\rho = 2$ in comparison with an average m value of about 60. For the prior on ϕ^2 , we used $a = 0.5$ and $d = 3$, again negligibly small compared to the sample information.

Four parallel chains were run and convergence was obtained for all parameters before 1000 iterations. After discarding the first 1000, 10000 sets of generated parameter values were kept, and Table 1 gives some posterior measures of η with AZT as baseline treatment.

Table 1. Posterior means of change-point parameters for the four treatments with 95% Highest Posterior Density Intervals.

	Treatment	Mean (Days)	95% HPD
η_1	AZT	341.03	246–434
$\eta_1 + \eta_2$	DDI	372.63	241–514
$\eta_1 + \eta_3$	AZT + DDI	419.48	263–591
$\eta_n + \eta_4$	AZT + DDC	406.02	275–548

From the large HPD intervals in Table 1 it is clear that there is no conclusive evidence of a treatment effect on the time from change-point to AIDS, so the model is simplified by replacing the mean $z_j\eta$ in equation (4.1) by a single parameter η , thus assuming a common mean change-point for all treatments. The model still allows for different slopes due to treatment effects. Measures of the marginal posteriors of some parameters in this case are given in Table 2.

Table 2. Posterior measures assuming a single population change-point

Parameter	Mean	95% HPD
η	381.36	311–455
ϕ^2	31472	15977–49893
σ^2	0.0566	0.0449–0.0684

The results indicate that the time from change-point to diagnosis of AIDS has a mean of 381 days and a standard deviation of 177 days. This is similar to the results obtained by Kiuchi et al. (1995), which gave a mean of 362 days with standard deviation of 257 days. The marginal posteriors of the elements of $M(3 \times 4)$ show no differences in slope before the change-point, but there are

significant differences in the rates of decrease after the change, with patients on AZT showing the fastest rate of decline. If the assumption of a common mean change-point is made, this may only mean that patients on AZT are diagnosed with AIDS at a lower CD4 cell count than patients on other treatments. This can be seen in Figure 2. However, the fact that the rates of decline differ may strengthen the evidence that the change-points are also different.

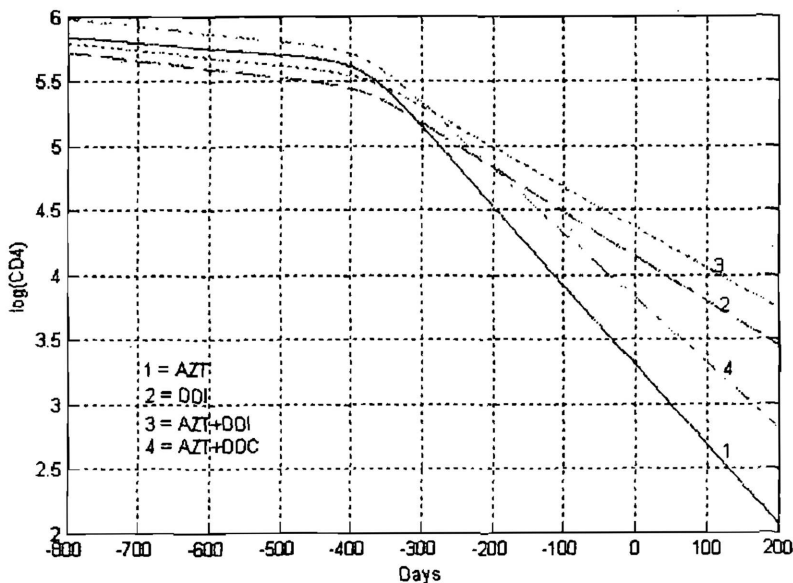


Figure 2. Mean regression lines for four treatments, assuming a common change-point. Origin at time of AIDS diagnosis.

The mean CD4 cell count over all treatments is about 270 at the change and drops to about 31 (for AZT) and 90 (for AZT+DDI) at time of diagnosis. The standard deviation of the predictive distribution of cell count at time of AIDS diagnosis for treatment AZT+DDI is 49. This large standard deviation shows

that cell count is a poor marker for AIDS, and patients in this study have been diagnosed with cell counts from as low as 3 to as high as 450. The mean slope of the linear segment after the change-point is on average, more than 5 times steeper than before.

Figure 3 shows the predictive survival function until diagnosis of AIDS for a future observation with the 95% credibility interval. Here the origin is at the time a change-point is observed. So there is a probability of 0.08 that a change will be observed only after diagnosis of AIDS, and a probability of 0.71 that a patient will stay AIDS-free for at least 200 days after a change.

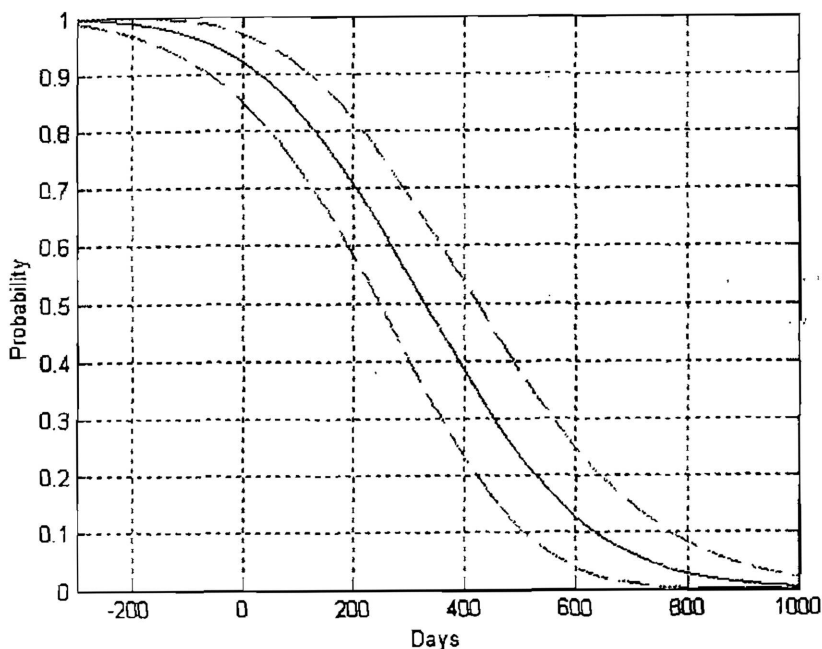


Figure 3. Predictive survival function of time to AIDS diagnosis, with 95% equal-tailed credibility interval. Origin at position of the change-point.

6. Discussion

In this paper a Bayesian model for analysing multiple sequences, each possibly containing a change-point, is proposed. An important point is that there is uncertainty about the existence of a change-point in each sequence. In the analysis the first step is to find the probability of a change-point in each sequence. This is done by means of the fractional Bayes factor of O'Hagan (1995). Next these probabilities are used as weights in deriving the posterior distribution of the population change-point.

There is compelling evidence in the data that in many cases (but not all) a sharp decline in CD4 cell count precedes the diagnosis of AIDS. Of the 32 uncensored sequences, 14 have posterior probabilities for a change-point of more than 0.8, while 9 have posterior probabilities of less than 0.5. A biological reason for the existence of a change-point could be the onset of antiviral resistance. The analysis shows little evidence of differences among treatments with respect to time from change-point to AIDS, but there are differences in the rates of decline after the change. Again one could hypothesise that the antiviral resistance is less effective in some cases, but without significantly prolonging time to AIDS.

The inclusion of censored observations is also essential. Since our main interest is in the distribution of the time from change-point to AIDS diagnosis, and patients entered the study at varying levels of infection, only those sequences that show some evidence of a change-point are included. The same analysis without the censored observations yielded a mean time to AIDS of 247 days compared to the final estimate of 381 days.

In this study we have used a log transformation on the cell counts to stabilise variances. This works well and also enables us to assume a common error variance over all sequences, as there is little variation among the estimated

error variances of the individual sequences. Kiuchi et al. (1995) also used the log transformation, while DeGruttola, Lange and Dafni (1991) and Lange, Carlin and Gelfand (1992) used square root transformations.

A simplification is the assumption that the rate of decline is the same, irrespective of whether the change-point occurs before or after the diagnosis of AIDS. It may be reasonable to cater for the possibility that the decline preceding AIDS may be different from decline following AIDS. However, in this data set there is very little information on rate of decline if the change-point is after AIDS diagnosis, the probability of that happening being estimated as 0.08 from Figure 3.

Some refinements to the model may be possible. In this paper the probability of a change-point in each sequence is calculated individually, ignoring the hierarchical model later adopted for the whole data set. It may be possible to use an iterative procedure, recalculating the probabilities using the proper distributions for the parameters obtained from the MCMC sampling as priors. The convergence properties of such an approach has, however, not yet been investigated.

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